

Project Initialization and Planning Phase

Date	July 7, 2026
Team ID	
Project Title	Machine Learning-Based Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project aims to develop an intelligent flood prediction and early warning system using machine learning techniques. The system analyzes historical weather data such as annual rainfall, monsoon rainfall, seasonal rainfall, and cloud visibility to predict the likelihood of flood occurrences. The solution helps disaster management authorities issue timely warnings, improve evacuation planning, and reduce the impact of floods on human life and property.

Project Overview	
Objective	Develop a machine learning-based system that predicts the probability of flood occurrence using historical and current weather parameters with high accuracy.
Scope	The project includes data collection, preprocessing, feature engineering, model training, evaluation, comparison of multiple algorithms, and deployment using Flask. The system is designed for real-time flood risk prediction and can be deployed on IBM Cloud for scalable access.
Problem Statement	
Description	Floods are among the most destructive natural disasters, causing loss of life, displacement of people, and significant economic damage. Existing forecasting methods often fail to provide timely and accurate warnings, leaving authorities with insufficient time to respond effectively.
Impact	An accurate flood prediction system can provide early warnings, support evacuation planning, improve disaster response, optimize resource allocation, and reduce the overall impact of flood-related disasters.
Proposed Solution	
Approach	Historical weather data is preprocessed and used to train multiple machine learning models including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost. The best-performing model is integrated into a Flask web application to provide real-time flood risk predictions.
Key Features	• Automated Flood Risk Prediction • Early Warning Support System • High Accuracy using Machine Learning • Real-Time Prediction through Web Application • User-Friendly Interface • Fast Decision-Making for Authorities • IBM Cloud Deployment Support • Scalable and Efficient Architecture

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Space for datasets, trained models, and application files	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, XGBoost
Development Environment	IDE	Jupyter Notebook, Visual Studio Code, PyCharm
Data		
Data	Source, Size, Format	Flood Prediction Dataset (Kaggle), CSV Format, Historical Weather and Rainfall Records